# Supplementary Material: Grounding Image Composition with Label-Conditioned Visual Evidence

## A Annotation Protocol and Annotators

Each KU-PCPE task showed one photograph and one positive KU-PCP composition label. The prompt asked annotators to select the segments that explain the queried composition. The same image could appear in several tasks when KU-PCP assigned it several labels. These tasks were completed separately.

The interface initially displayed the image and the queried label. Clicking in the image revealed precomputed candidate regions; a preview showed the current mask (Figure 1a). Annotators could select several candidates, add a new region, split or remove a region, refine its boundary, undo changes, and confirm the final union. After confirmation, they could compare the region mask with an automatically detected semantic-line overlay (Figure 1b) and record a soft segment-versus-line rating. This rating did not alter the binary mask.

The three test experts worked independently and did not see one another's masks. All had formal education in arts or design and more than five years of professional experience in art, design, photography, or professional visual practice. A separate annotator completed the training split. The train and test annotations used the same image-label task, interface, and class guide. The training masks provide model supervision; they are not used to estimate expert agreement.

All scheduled tasks concerned labels already present in KU-PCP. They were not turned into negative examples when an image was difficult. The released set contains only completed tasks, and no released mask is empty. Before consensus construction, exports were checked for complete task coverage, readable masks, correct image dimensions, and nonempty selections. Flagged masks were reviewed and corrected before the final test set was formed.

The preserved annotation caches identify SAM2 Hiera-L with configuration `configs/sam2/sam2_hiera_l.yaml` and checkpoint `sam2_hiera_large_072824.pt`. Cached candidates retain their original ID and score ordering and allow exact replay. The historical generation thresholds were not stored in those records. The benchmark uses the separately specified SAM2.1 bank in Supplement H.1.

## B Class Guide and Binary Mask Interpretation

### B.1 Guide Used During Collection

The definitions and "what to look for" cues in Table 1 reproduce the guide given to annotators, with only the layout condensed. Section C provides separate examples selected from the benchmark.

Table 1: Composition definitions and visual cues given to annotators.

| Class | Definition | What to look for |
|---|---|---|
| Rule-of-Thirds (RoT) | Main subject placed along a rule-of-thirds location, especially near a vertical trisecting line rather than dead center. | Off-center subject placement; alignment near thirds-grid intersections or vertical thirds lines; balance created by asymmetrical composition. |
| Vertical | Strong repeated or dominant vertical elements emphasizing height or upright structure. | Dominant upright lines or repeated vertical rhythm; height-emphasizing composition; vertical structure outweighs horizontal flow. |
| Horizontal | Strong horizontal structure or leading line, often horizon-like or calm lateral division. | Dominant left-to-right directional structure; horizon-like or layered horizontal bands; lateral organization more important than vertical cues. |
| Diagonal | Dominant slanted line or directional structure that creates diagonal flow. | Strong angled composition direction; motion or tension created by slanted alignment; diagonal path that guides gaze. |

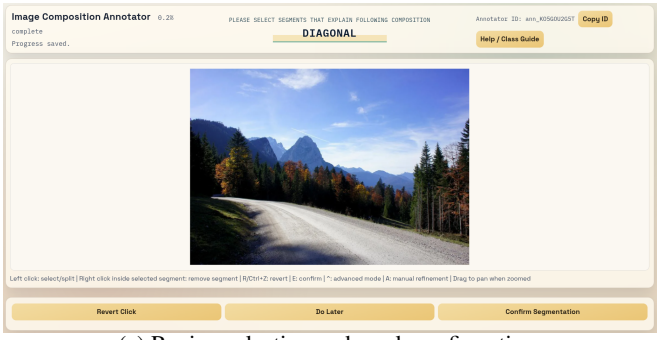
| Class | Definition | What to look for |
|---|---|---|
| Curved | Dominant curved flow or path guiding the eye. | Arc-like or S-shaped compositional path; curve-driven eye guidance across the frame; smooth curvature dominates structural reading. |
| Triangle | Triangular or implied triangular arrangement, including vanishing-line based triangles. | Three-point or wedge-like structural arrangement; implied triangular balance in key regions; triangle formed by lines, edges, or grouped elements. |
| Center | Main subject placed at or very near the image center. | Dominant visual mass around image midpoint; supporting structure reinforcing central placement; minimal directional pull away from center. |
| Symmetric | Mirrored structure around a dominant axis; here this is mainly horizontal symmetry. | Clear mirrored balance around one axis; matching structural regions on both sides; axis-driven composition dominates interpretation. |
| Pattern | Repetition of similar shapes or structures creating a patterned visual effect. | Repeating visual units; regularity or rhythmic repetition; pattern itself acts as composition driver. |

### B.2 Binary Mask Interpretation

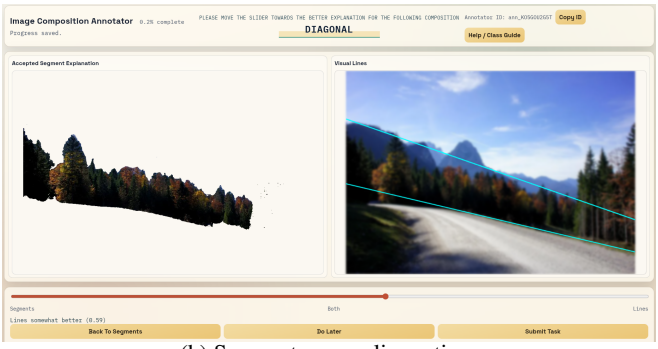
The guide identifies the composition to look for. The binary annotation is the union of visible image regions selected as explaining that composition. It does not impose one object granularity: a valid union may contain an object, an object part, several disconnected supports, background, or negative space. Words such as line, path, and axis describe the relation being explained; the target is the visible carrier or partition, not a synthetic guideline. When the initial candidates did not express the intended support, the interface allowed additions, splits, removals, and manual refinement. Table 2 explains how the same type of mask can represent these different compositions and where alternative selections arise.

## C Examples Across All Classes

Figures 2 to 4 show one high-agreement task and one ambiguous task for every class. These are selected from the released expert test set and are benchmark examples, not reconstructed collection-time guide images. All masks use the same blue overlay and boundary style as the main paper. Table 3 identifies what the experts interpret differently in each ambiguous example, from opposite sides of a boundary to different extents of the same subject.




(a) Region selection and mask confirmation.




(b) Segment-versus-line rating.

Figure 1: The two annotation states. Candidate regions support efficient mask editing; the semantic-line overlay is a comparison aid rather than a target.

| Class | How the mask can express the composition | Common ambiguity |
|---|---|---|
| Center | Centrally placed subject, part, or supporting central region. | Main subject alone versus a larger central structure. |
| Rule of thirds | Off-center subject or region whose placement creates the thirds relation. | Subject boundary versus surrounding context; left versus right placement. |
| Horizontal | Regions or elements forming a lateral division or arrangement. | Upper versus lower side of a complementary partition. |
| Vertical | Dominant or repeated upright supports. | Complete structures versus their most salient vertical parts. |
| Diagonal | One elongated carrier or several aligned elements creating slanted flow. | One side of a diagonal partition versus the other. |
| Symmetric | Corresponding reflected regions and, where relevant, their central support. | One side, both sides, or the region joining them. |
| Curved | One carrier or several elements jointly forming a perceptual curve. | The curve carrier versus a broader enclosing region. |
| Triangle | Objects, parts, surfaces, or grouped elements forming an explicit or implied triangle. | Triangle-forming elements versus the enclosed region. |
| Pattern | Several repeated instances whose regularity drives the composition. | A representative subset versus the full repeated field. |

Table 2: How the common region representation applies to the nine classes. These descriptions clarify the mask representation and were not additional collection instructions.

| Class | Representational issue illustrated by the ambiguous row |
|---|---|
| Horizontal | Complementary sides of a lateral partition and an intersecting subject. |
| Vertical | Central upright passage versus surrounding upright walls. |
| Diagonal | Opposite image regions separated by the same slanted boundary. |
| Curved | Water, land, or sky regions that jointly make the curved path visible. |
| Triangle | Triangle-forming architectural regions versus the broader enclosed shape. |
| Rule of thirds | Main subject alone versus subject with nearby supporting context. |
| Center | Central architectural mass versus a smaller central portion. |
| Symmetric | Paired reflected regions versus the central path and its surroundings. |
| Pattern | A subset of repeated units versus the larger repeated field. |

Table 3: The ambiguity represented in each difficult example.

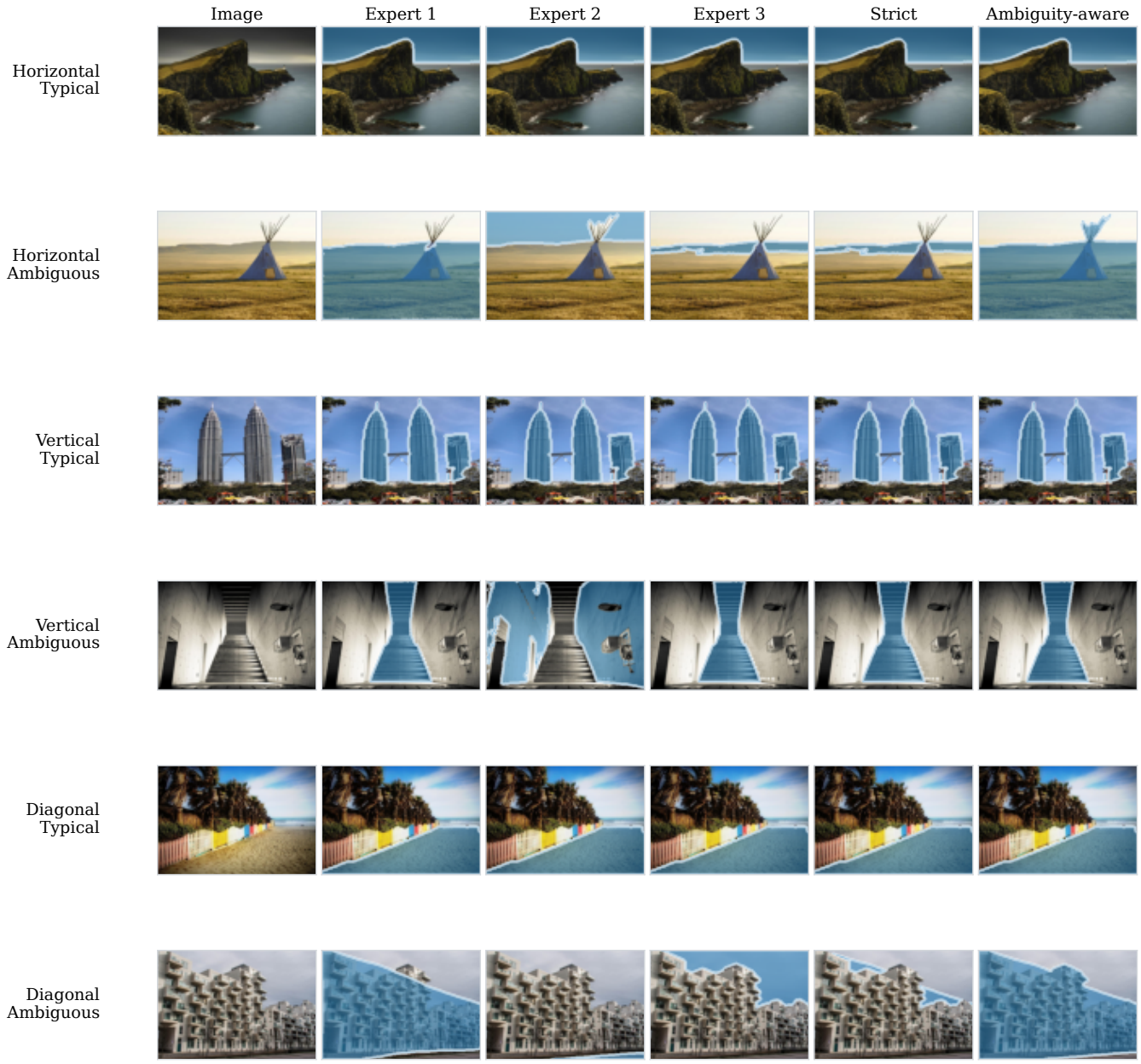



Figure 2: Typical and ambiguous examples for horizontal, vertical, and diagonal. The ambiguous rows show complementary partitions or disagreement about which visible carrier best expresses the direction.

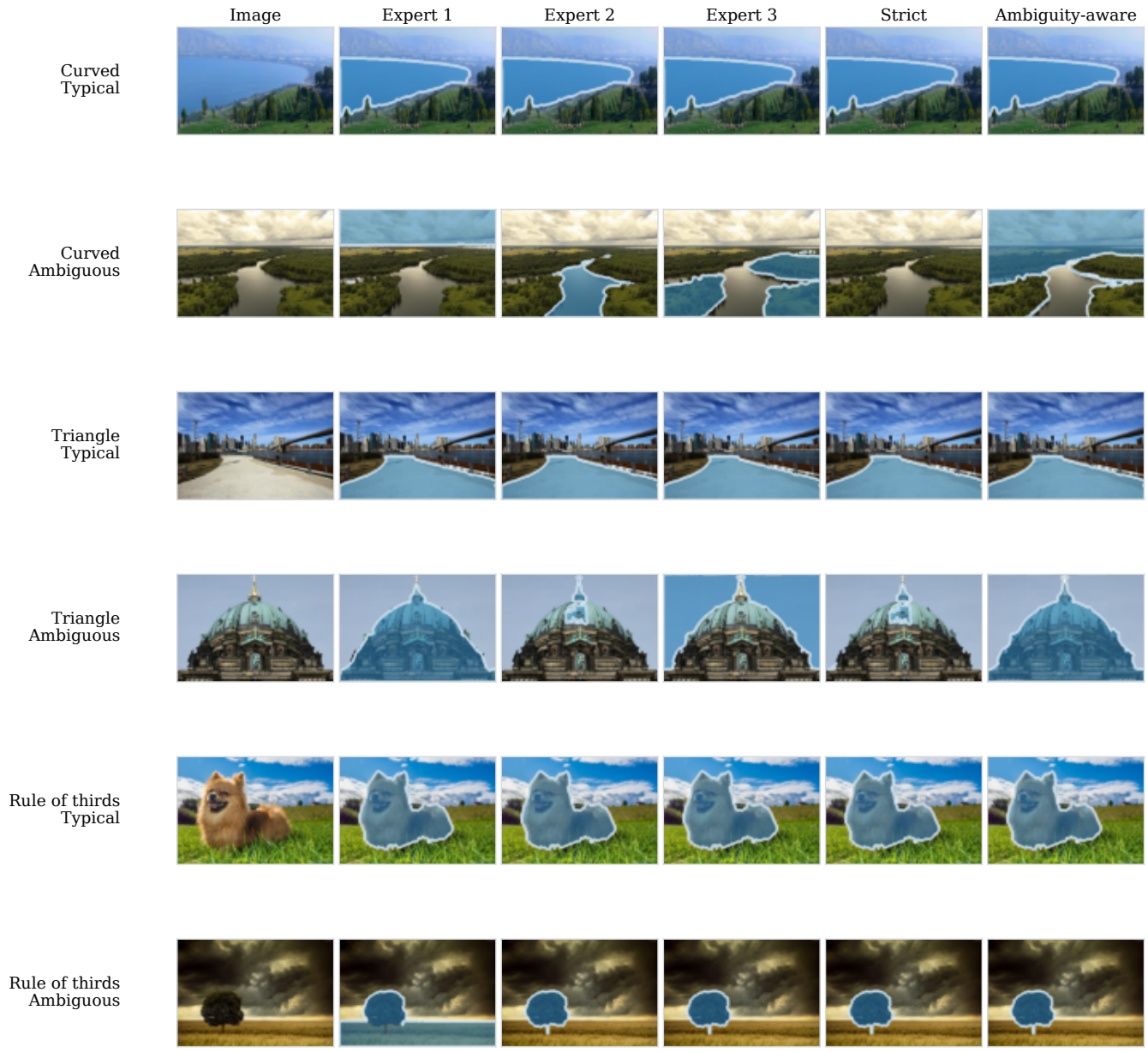



Figure 3: Typical and ambiguous examples for curved, triangle, and rule of thirds. Ambiguity concerns the curve carrier, triangle-forming regions versus their interior, and subject extent or supporting context.

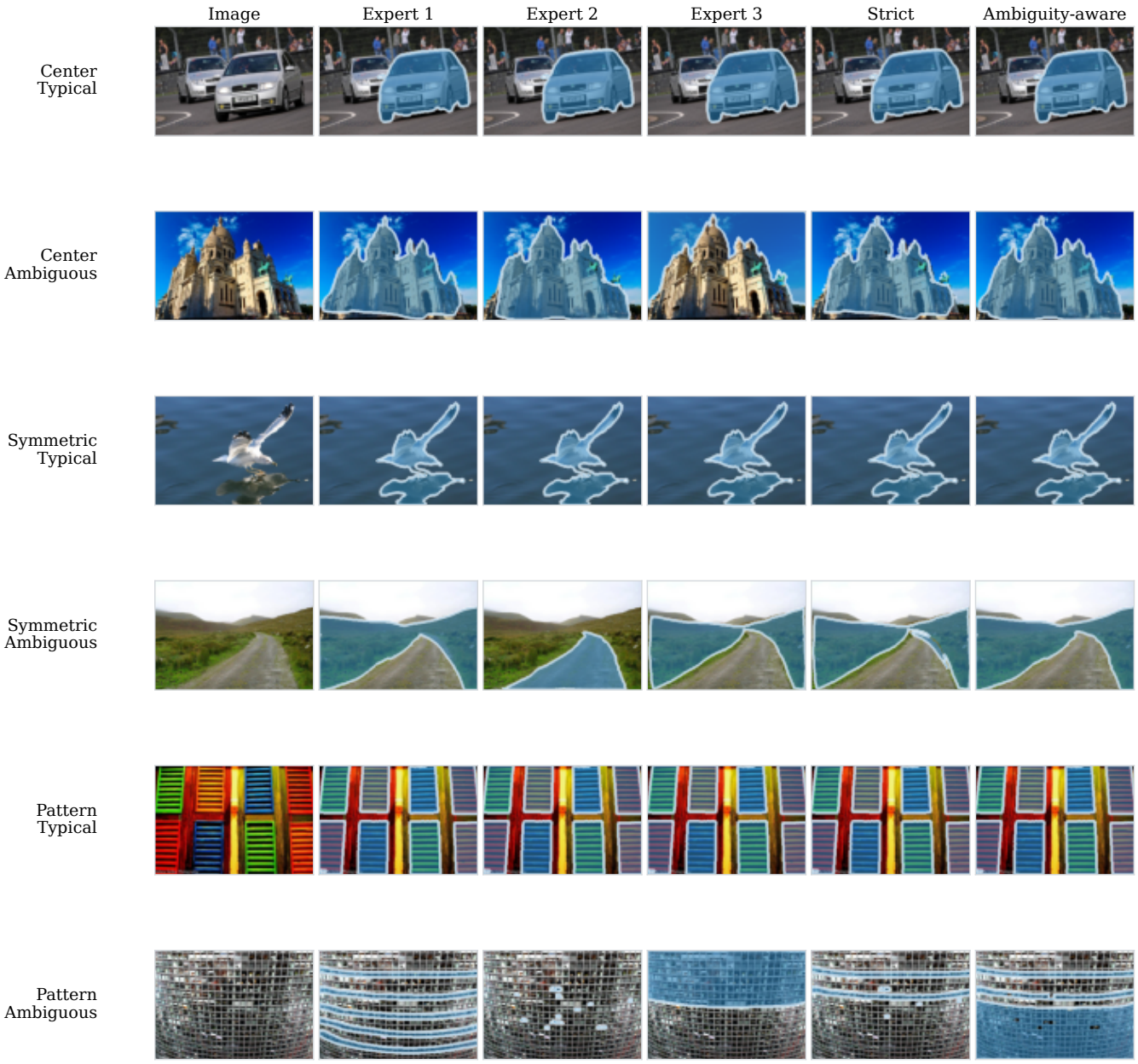



Figure 4: Typical and ambiguous examples for center, symmetric, and pattern. The difficult rows show variation in subject extent, paired versus surrounding support, and how many repeated instances to include.

| Metric | Mean | 95% task-bootstrap CI |
|---|---|---|
| Dice | .8666 | [.8577, .8750] |
| IoU | .8054 | [.7939, .8167] |
| Boundary F1 | .7541 | [.7411, .7669] |
| Overcoverage | .1131 | [.1058, .1203] |
| Hit@0.5 | .9312 | [.9209, .9410] |

Table 4: Held-out expert agreement with the aligned consensus of the other two experts.

| Class | $n$ | Area | Pair D | Pair I | B-F1 | Maj. D | $\kappa$ | Flip |
|---|---|---|---|---|---|---|---|---|
| Horizontal | 211 | .3882 | .6371 | .6241 | .7320 | .9539 | .8281 | .5118 |
| Vertical | 66 | .2426 | .6557 | .5599 | .6619 | .8362 | .5782 | .1515 |
| Diagonal | 189 | .3741 | .5973 | .5566 | .6311 | .8805 | .5433 | .4656 |
| Curved | 92 | .2961 | .6105 | .5593 | .6103 | .8556 | .5240 | .3696 |
| Triangle | 171 | .3126 | .8332 | .7860 | .7611 | .9351 | .7796 | .1170 |
| Rule of thirds | 125 | .0939 | .9538 | .9326 | .9473 | .9782 | .9426 | .0080 |
| Center | 311 | .2051 | .8427 | .8049 | .8008 | .9235 | .8014 | .0354 |
| Symmetric | 99 | .4114 | .8239 | .7419 | .5253 | .9165 | .7152 | .0505 |
| Pattern | 63 | .4794 | .5324 | .4224 | .5729 | .8092 | .2510 | .5397 |

Table 5: Agreement by class. Pair D and Pair I are raw pairwise Dice and IoU; Maj. D is Dice to the aligned majority; Flip is the fraction of tasks with at least one complement alignment.

# D Consensus and Agreement

The descriptive agreement table (Table 5) jointly chooses the complement assignments of experts 2 and 3 to maximize mean pairwise Dice, keeping expert 1 fixed. Its aligned kappa, majority Dice, and flip rate use that descriptive convention. The benchmark targets use the reference-based rule below; these two constructions differ on 48 of the 1,327 tasks.

Let $M_1, M_2, M_3$ be the three expert masks for one image-label task. The strict target is their pixelwise majority. The ambiguity-aware target uses $M_1$ as a reference orientation. For $j \in \{2, 3\}$, it replaces $M_j$ by its complement only when $\mathrm{Dice}(\neg M_j, M_1) > \mathrm{Dice}(M_j, M_1)$, then takes the pixelwise majority. The rule is deterministic and local to each task. It handles complementary choices around the same partition; it does not assume complement equivalence for every mask.

In pseudocode:

```
strict = majority(M1, M2, M3)
aligned = [M1]
for M in [M2, M3]:
  if Dice(not M, M1) > Dice(M, M1):  append not M
  else:  append M
ambiguity-aware = majority(aligned)
```

For held-out expert evaluation, the other two masks are aligned by the same rule and their pixelwise mean is thresholded at .5. With two binary masks this is their union. The held-out mask is also complemented only if doing so improves Dice to the peer target. Metrics are averaged over the three held-out choices within each task. Confidence intervals then resample the 1,327 tasks, so the three experts are not treated as independent examples. Table 4 summarizes this reference; the high overlap indicates substantial agreement after accounting for complementary choices. Figure 5 distinguishes those choices from disagreement about which regions matter. Rule of thirds, center, and triangle have particularly high raw agreement (Table 5).

## D.1 Strict Results

## D.2 Ambiguity-Aware Results

Tables 6 and 7 show that changing the consensus rule preserves the broad ordering of methods, with SegFormer-B0 remaining the strongest deployable segmenter. Oracle@3 is optimized independently for each target because it is a target-aware greedy coverage reference rather than a guaranteed optimum. Every deployable prediction is fixed and only rescored across the two views.

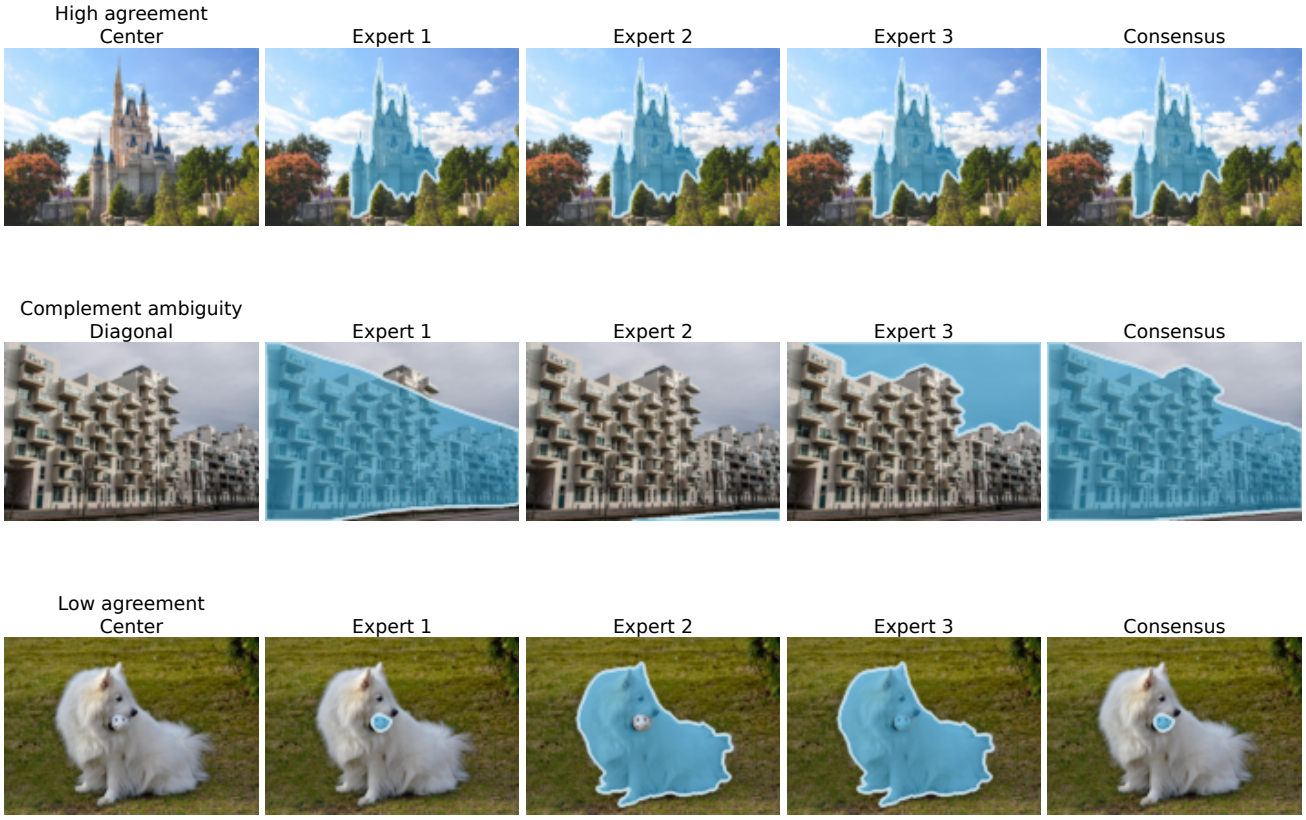



Figure 5: High agreement, complementary ambiguity, and low agreement. The middle row illustrates how opposite sides of one partition can align in the ambiguity-aware view.

| Method | Dice | IoU | B-F1 | Overcoverage | Hit@0.5 |
|---|---|---|---|---|---|
| Random matched area | .2250 [.2191, .2310] | .1312 [.1273, .1350] | .0357 [.0319, .0397] | .7463 [.7371, .7556] | .0015 [.0000, .0038] |
| Train class template | .4512 [.4387, .4635] | .3186 [.3085, .3287] | .1621 [.1555, .1689] | .5104 [.4943, .5268] | .4891 [.4627, .5170] |
| SAM2.1 Oracle@3 | .6786 [.6591, .6979] | .6159 [.5951, .6368] | .6486 [.6298, .6681] | .0838 [.0724, .0956] | .6714 [.6458, .6963] |
| SAM2.1 Template@3 | .4131 [.3941, .4318] | .3341 [.3160, .3521] | .4429 [.4268, .4593] | .5319 [.5112, .5527] | .3956 [.3693, .4213] |
| U-Net, no label | .4821 [.4662, .4980] | .3691 [.3545, .3835] | .3029 [.2921, .3138] | .4962 [.4787, .5140] | .5200 [.4928, .5463] |
| U-Net + label | .5870 [.5715, .6024] | .4725 [.4574, .4877] | .3864 [.3742, .3986] | .4191 [.4021, .4361] | .6511 [.6255, .6767] |
| SegFormer-B0 | .6624 [.6441, .6801] | .5767 [.5586, .5947] | .4335 [.4193, .4481] | .3197 [.3017, .3377] | .7362 [.7129, .7596] |
| Qwen3-VL-32B | .3988 [.3792, .4186] | .3291 [.3102, .3480] | .4383 [.4217, .4550] | .5380 [.5160, .5601] | .3828 [.3564, .4092] |

Table 6: Results against the strict consensus. Entries are means with 95% task-bootstrap intervals.

| Method | Dice | IoU | B-F1 | Overcoverage | Hit@0.5 |
|---|---|---|---|---|---|
| Random matched area | .2375 [.2315, .2436] | .1394 [.1354, .1433] | .0370 [.0332, .0409] | .7173 [.7068, .7275] | .0068 [.0030, .0113] |
| Train class template | .4617 [.4502, .4734] | .3254 [.3157, .3351] | .1691 [.1624, .1760] | .4815 [.4655, .4975] | .4936 [.4665, .5200] |
| SAM2.1 Oracle@3 | .6718 [.6533, .6903] | .6054 [.5846, .6263] | .6317 [.6124, .6505] | .0674 [.0575, .0776] | .6609 [.6353, .6865] |
| SAM2.1 Template@3 | .4147 [.3963, .4337] | .3337 [.3160, .3517] | .4475 [.4317, .4629] | .5104 [.4895, .5310] | .3986 [.3730, .4258] |
| U-Net, no label | .4958 [.4803, .5115] | .3790 [.3650, .3930] | .3071 [.2964, .3183] | .4585 [.4410, .4757] | .5335 [.5079, .5607] |
| U-Net + label | .5970 [.5820, .6120] | .4791 [.4645, .4939] | .3904 [.3788, .4025] | .3903 [.3739, .4070] | .6609 [.6353, .6850] |
| SegFormer-B0 | .6636 [.6461, .6813] | .5762 [.5582, .5937] | .4368 [.4228, .4511] | .3000 [.2826, .3175] | .7340 [.7106, .7573] |
| Qwen3-VL-32B | .3946 [.3752, .4144] | .3243 [.3060, .3434] | .4396 [.4235, .4555] | .5231 [.5013, .5446] | .3760 [.3512, .4024] |

Table 7: Results against the ambiguity-aware consensus. Entries are means with 95% task-bootstrap intervals.

| Source | Area fraction | Components | Boundary complexity | Modality score |
|---|---|---|---|---|
| Train annotator | .2436 [.2381, .2490] | 10.27 [9.55, 11.05] | 9.71 [9.50, 9.91] | .1955 [.1880, .2031] |
| Expert 1 | .2462 [.2368, .2555] | 7.96 [6.98, 9.05] | 8.69 [8.40, 9.00] | .3373 [.3234, .3512] |
| Expert 2 | .2520 [.2426, .2615] | 8.61 [7.26, 10.38] | 8.38 [8.12, 8.65] | .2499 [.2355, .2646] |
| Expert 3 | .2863 [.2767, .2960] | 7.92 [7.09, 8.82] | 9.15 [8.85, 9.47] | .4949 [.4793, .5104] |
| Strict consensus | .2537 [.2444, .2632] | 9.05 [8.12, 10.05] | 8.94 [8.65, 9.23] | .3607 [.3480, .3733] |
| Ambiguity-aware | .2827 [.2722, .2928] | 8.77 [7.83, 9.81] | 8.82 [8.54, 9.12] | .3607 [.3477, .3736] |

Table 8: Mask statistics for the training annotator, each test expert, and the two consensus targets. Values are means with 95% task-bootstrap intervals.

| Method | Expert 1 | Expert 2 | Expert 3 | Strict | Ambig. |
|---|---|---|---|---|---|
| Random matched area | .2199 | .2220 | .2419 | .2250 | .2375 |
| Train class template | .4433 | .4471 | .4444 | .4512 | .4617 |
| SAM2.1 Oracle@3 | .6742 | .6948 | .6688 | .6786 | .6718 |
| SAM2.1 Template@3 | .3957 | .4242 | .4006 | .4131 | .4147 |
| U-Net, no label | .4658 | .4545 | .4951 | .4821 | .4958 |
| U-Net + label | .5649 | .5770 | .5867 | .5870 | .5970 |
| SegFormer-B0 | .6323 | .6483 | .6418 | .6624 | .6636 |
| Qwen3-VL-32B | .3775 | .4074 | .3986 | .3988 | .3946 |

Table 9: Dice after scoring each method against the three individual experts and both consensus targets. Oracle@3 is selected separately for each target; all other predictions are fixed.

# E Variation Across Annotations

The train and test image splits differ, and the training split has one annotator while the test split has three experts. The comparisons below describe the supervision and evaluation targets without trying to separate image differences from annotator differences. Components are 8-connected foreground components. Boundary complexity is contour perimeter divided by the square root of foreground area. Intervals bootstrap image-label tasks.

Table 8 shows differences in mask area and component structure across annotators and targets. Consensus construction also changes area and component structure, particularly when complementary partitions are aligned. This is why training masks are treated as supervision rather than a human-reliability sample.

Table 9 shows the same broad ordering for all five targets: SegFormer-B0 is the strongest deployable segmenter, Qwen3-VL-32B remains below full-bank Template@3, and Oracle@3 remains a reference-dependent coverage control. The model comparison is therefore not an artifact of one expert or one consensus construction.

| Property | KU-PCP | KU-PCPE |
|---|---|---|
| Primary supervision | Image-level composition labels | Image-label region evidence masks |
| Localized output | Class-specific boxes, lines or axes, triangles, and regions produced by automatic methods | Binary regions selected by people |
| Representation across classes | Different geometric forms for different classes | One mask representation for all nine classes |
| Test interpretations | One output for each supported detector | Three expert masks and two consensus views |
| Intended use | Composition classification and element detection | Conditional grounding of human-selected evidence |

Table 10: Relationship between the two representations.

| Target | $n$ | Line in target | +5 px | Target covered | Line D | Filled D | Filled I |
|---|---|---|---|---|---|---|---|
| Ambiguity-aware | 171 | .398 [.354, .442] | .489 [.441, .536] | .036 | .064 | .585 [.544, .624] | .462 [.422, .501] |
| Strict | 171 | .374 [.332, .415] | .465 [.419, .511] | .036 | .065 | .572 [.530, .612] | .451 [.411, .490] |

Table 11: Comparison with KU-PCP triangle detector output. D and I denote Dice and IoU; brackets give 95% task-bootstrap intervals where reported.

# F Relationship to Original KU-PCP Elements

KU-PCP defines nine image-level composition classes and proposes class-specific automatic schemes for locating dominant geometric elements, including subject boxes, leading lines, a symmetry axis, triangles, and sky regions. These outputs have different geometric forms. KU-PCPE uses one human-selected binary region representation for every positive image-label task and provides three interpretations on the test split. Its contribution is therefore not the first localization of any composition cue, but a common human grounding target suitable for agreement analysis and cross-class evaluation.

Table 10 contrasts the class-specific geometric outputs with the regions selected by people. Because boxes, lines, axes, triangles, and regions do not share one natural mask conversion, a common cross-class overlap score is not defined. We report the 171 triangle tasks for which a matched detector output can be evaluated without replacing the original method with another detector.

The KU-PCP triangle output stores two detected sides as four endpoints. We rasterize the sides at the method's 10-pixel width. "Line in target" is the fraction of side pixels inside the KU-PCPE target; the tolerant version dilates the target by five pixels at the detector resolution. "Target covered" is the fraction of KU-PCPE pixels occupied by the sides. To compare regions, we also fill the triangle induced by the two base points and the mean of the two apex endpoints. This filled region is reported separately from the native side output.

In Table 11, the native sides overlap part of the selected human support but cover only 3.6% of the region target. Filling the triangle raises Dice to .585. This shows a meaningful relationship to the region organized by the primitive, but not equivalence between a thin geometric output and the visible regions people select. Figure 6 illustrates this distinction: triangle-forming objects and the enclosed image region need not coincide.

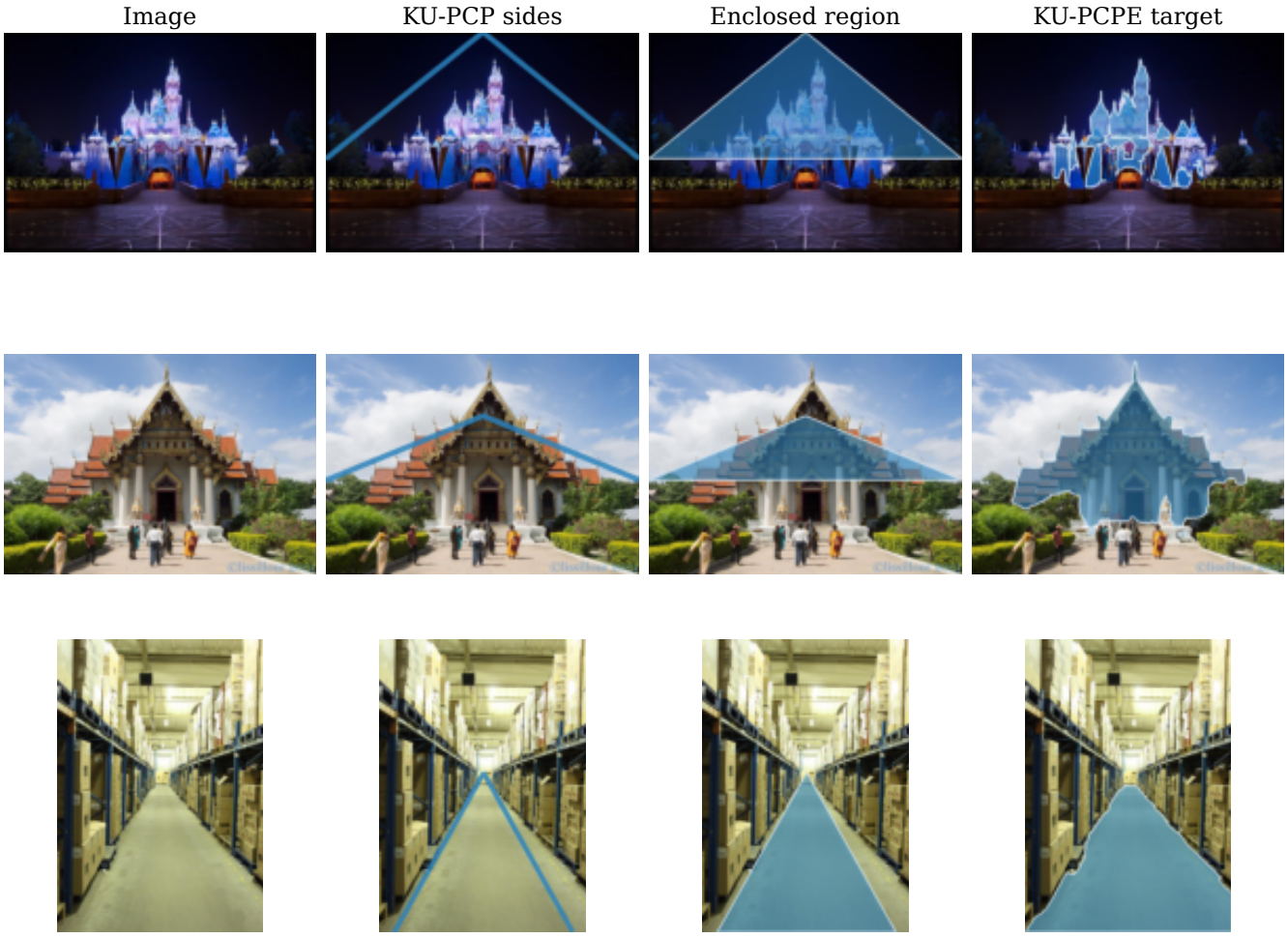



Figure 6: Matched triangle examples across the observed overlap range. Columns show the photograph, KU-PCP's detected sides, the region enclosed by those sides, and the KU-PCPE ambiguity-aware target.

# G Results by Annotation Workflow

The complete export provides retained interaction records for the training annotator and all three test experts. Process percentages use annotations as the denominator: 4,093 training and 3,981 test records. Workflow comparisons instead use 1,327 unique test image-label tasks, each counted once. A task is *no edit logged* when none of its three experts recorded a geometry action, and *edit logged* when at least one expert did.

Geometry actions are dynamic region addition, region acceptance, positive or negative points, applying or undoing manual refinement, splitting, removing a region, or clearing the current region. Manual tool use includes entering, applying, exiting, or undoing refinement. These definitions preserve the original event categories. Event counts and type presence are recoverable from the released timelines despite their five-second time bins.

The groups describe retained interface records, not complete revision histories. Discarded drafts are unavailable, and 71 symmetric masks were replaced after the earlier logged pass. Agreement uses the final three expert masks; model evaluation uses the final ambiguity-aware consensus. Neither predictions nor oracle masks are selected anew within a group. The comparison measures an association with task difficulty, not the effect of editing.

| Annotation | $n$ | Raw pair D | Comp. pair D | B-F1 | Pair $\kappa$ |
|---|---|---|---|---|---|
| No edit logged | 615 | .8579 [.8384, .8773] | .9132 [.9004, .9257] | .8858 [.8700, .9012] | .8072 [.7792, .8345] |
| Edit logged | 712 | .6434 [.6218, .6652] | .7467 [.7295, .7634] | .5822 [.5631, .6011] | .5292 [.4993, .5589] |

Table 12: Expert agreement by all-expert workflow group. Brackets give 95% task-bootstrap intervals. Comp. pair D preserves the descriptive pairwise convention: for each pair in the published expert order, complement the second mask only when Dice improves, then average over the three pairs. It differs from the joint alignment used for majority agreement.

| Method | Dice | IoU | B-F1 | Overcoverage | Hit@0.5 |
|---|---|---|---|---|---|
| Random matched area | .215 [.207, .224] | .125 [.120, .131] | .032 [.026, .037] | .754 [.739, .767] | .008 [.002, .016] |
| Train class template | .485 [.468, .501] | .346 [.331, .360] | .167 [.157, .178] | .478 [.456, .500] | .535 [.496, .574] |
| SAM2.1 Oracle@3 | .785 [.759, .810] | .739 [.710, .767] | .769 [.743, .793] | .025 [.017, .034] | .769 [.735, .802] |
| SAM2.1 Template@3 | .481 [.453, .509] | .397 [.369, .426] | .535 [.513, .557] | .482 [.452, .512] | .465 [.426, .504] |
| U-Net, no label | .558 [.535, .581] | .444 [.422, .466] | .361 [.343, .379] | .431 [.405, .456] | .620 [.582, .659] |
| U-Net + label | .671 [.652, .692] | .556 [.535, .576] | .457 [.439, .475] | .330 [.308, .353] | .767 [.733, .800] |
| SegFormer-B0 | .766 [.743, .788] | .685 [.662, .708] | .524 [.504, .544] | .229 [.206, .252] | .855 [.826, .883] |
| Qwen3-VL-32B | .472 [.443, .502] | .401 [.372, .431] | .535 [.511, .559] | .486 [.455, .518] | .462 [.423, .501] |

Table 13: Results for the 615 tasks with no geometry action recorded by any expert. Brackets give 95% task-bootstrap intervals.

| Method | Dice | IoU | B-F1 | Overcoverage | Hit@0.5 |
|---|---|---|---|---|---|
| Random matched area | .257 [.249, .265] | .152 [.146, .157] | .042 [.036, .047] | .686 [.671, .701] | .006 [.001, .011] |
| Train class template | .442 [.426, .457] | .308 [.294, .321] | .171 [.162, .180] | .484 [.461, .507] | .458 [.420, .494] |
| SAM2.1 Oracle@3 | .574 [.548, .599] | .490 [.464, .516] | .513 [.489, .539] | .104 [.088, .121] | .567 [.529, .604] |
| SAM2.1 Template@3 | .358 [.333, .382] | .279 [.256, .301] | .372 [.352, .393] | .535 [.506, .564] | .341 [.306, .375] |
| U-Net, no label | .442 [.423, .461] | .323 [.305, .339] | .261 [.249, .273] | .482 [.459, .505] | .459 [.423, .496] |
| U-Net + label | .533 [.512, .552] | .413 [.393, .432] | .333 [.319, .348] | .442 [.419, .465] | .569 [.532, .605] |
| SegFormer-B0 | .575 [.550, .600] | .482 [.458, .507] | .361 [.343, .379] | .361 [.337, .387] | .629 [.594, .663] |
| Qwen3-VL-32B | .328 [.303, .352] | .258 [.236, .282] | .357 [.337, .378] | .555 [.525, .585] | .302 [.268, .336] |

Table 14: Results for the 712 tasks with at least one geometry action recorded by at least one expert. Brackets give 95% task-bootstrap intervals.

| Annotation | Class | $n$ | Pair D | O@3 | T@3 | SegF. | Qwen |
|---|---|---|---|---|---|---|---|
| No edit logged | Center | 203 | .912 | .776 | .548 | .828 | .509 |
| | Curved | 24 | .822 | .593 | .330 | .552 | .413 |
| | Diagonal | 61 | .760 | .780 | .477 | .534 | .512 |
| | Horizontal | 94 | .684 | .917 | .580 | .705 | .613 |
| | Pattern | 26 | .613 | .319 | .245 | .593 | .215 |
| | Rule of thirds | 103 | .977 | .915 | .368 | .896 | .393 |
| | Symmetric | 1 | 1.000 | .995 | .170 | .874 | .170 |
| | Triangle | 84 | .952 | .686 | .494 | .836 | .350 |
| | Vertical | 19 | .770 | .854 | .364 | .635 | .664 |
| Edit logged | Center | 108 | .713 | .591 | .377 | .565 | .300 |
| | Curved | 68 | .536 | .526 | .285 | .489 | .234 |
| | Diagonal | 128 | .520 | .529 | .299 | .414 | .281 |
| | Horizontal | 117 | .600 | .791 | .558 | .599 | .537 |
| | Pattern | 37 | .476 | .285 | .170 | .352 | .163 |
| | Rule of thirds | 22 | .844 | .719 | .203 | .753 | .220 |
| | Symmetric | 98 | .822 | .458 | .318 | .776 | .303 |
| | Triangle | 87 | .718 | .507 | .385 | .734 | .254 |
| | Vertical | 47 | .610 | .712 | .334 | .485 | .496 |

Table 15: Per-class Dice in the two all-expert workflow groups. Abbreviations follow the main paper. Only one symmetric task has no edit logged, so that class cannot support a reliable workflow comparison.

Table 12 shows lower agreement when any expert recorded an edit. Comparing Tables 13 and 14 shows lower scores for all four methods in that group. In Table 15, raw agreement, Oracle@3, and SegFormer are lower in the edit group in all nine classes; Template@3 and Qwen are lower in eight. The exception for the latter two is symmetric, whose no-edit group has one task. Giving each class equal weight yields no-edit/edit Dice of .832/.649 for raw agreement, .759/.569 for Oracle@3, .397/.325 for Template@3, .717/.574 for SegFormer, and .427/.310 for Qwen. These descriptive averages include the sparse symmetric group. The direction also holds when each expert's own records define the groups separately (Table 16). Table 17 shows that every annotator used geometry tools, although most annotations had no geometry action recorded. Manual tool use and other actions vary across annotators.

| Rater | Annotation | $n$ | Pair D | O@3 | T@3 | SegF. | Qwen |
|---|---|---|---|---|---|---|---|
| rater_01 | No edit logged | 859 | .793 [.774, .813] | .753 [.731, .775] | .467 [.443, .491] | .709 [.688, .729] | .453 [.428, .478] |
| rater_01 | Edit logged | 468 | .650 [.624, .676] | .523 [.494, .552] | .319 [.291, .347] | .580 [.550, .610] | .287 [.259, .316] |
| rater_03 | No edit logged | 770 | .817 [.797, .836] | .767 [.744, .791] | .482 [.456, .507] | .718 [.696, .740] | .462 [.435, .489] |
| rater_03 | Edit logged | 557 | .640 [.615, .664] | .540 [.512, .568] | .322 [.296, .349] | .588 [.560, .615] | .302 [.275, .329] |
| rater_04 | No edit logged | 846 | .807 [.788, .825] | .773 [.751, .795] | .484 [.459, .507] | .718 [.697, .739] | .469 [.444, .494] |
| rater_04 | Edit logged | 481 | .631 [.603, .657] | .493 [.465, .521] | .294 [.267, .320] | .568 [.538, .598] | .264 [.238, .291] |

Table 16: Sensitivity to the expert whose records define the groups. Each pair of rows partitions all 1,327 tasks using one expert; agreement still uses all three final masks and all models use the same consensus. These are three alternative groupings, not independent cohorts. Brackets give 95% task-bootstrap intervals.

| Rater | Split | Annotations | No edit | Edit | Manual | Split | Dynamic | Median actions |
|---|---|---|---|---|---|---|---|---|
| rater_02 | Train | 4,093 | 51.8% | 48.2% | 15.1% | 5.7% | 31.9% | 15 |
| rater_01 | Test | 1,327 | 64.7% | 35.3% | 16.5% | 7.5% | 27.4% | 12 |
| rater_03 | Test | 1,327 | 58.0% | 42.0% | 20.0% | 6.2% | 30.7% | 8 |
| rater_04 | Test | 1,327 | 63.8% | 36.2% | 16.1% | 4.7% | 28.3% | 6 |

Table 17: Interface use for every completed annotator. Percentages refer to annotations, not unique images. No edit/edit use the geometry-action definition above; manual is manual tool use and dynamic is dynamic region addition. Public rater IDs are pseudonyms. The test annotators each completed the same 1,327 tasks.

# H Full Benchmark Protocols and Results

## H.1 Proposal Generation and Selectors

Benchmark proposals are generated once per test image with SAM2.1 Hiera-L and the official `sam2.1_hiera_large.pt` checkpoint and `sam2.1_hiera_l.yaml` configuration. Automatic mask generation uses 32 points per side, 128 points per batch, predicted intersection-over-union (IoU) threshold .70, stability threshold .92, stability offset .70, one crop layer, box-NMS threshold .70, crop-point downscale factor 2, minimum region area 25 pixels, and mask-to-mask refinement. We retain at most 128 masks, ordered by predicted IoU and then area. The bank averages 54.48 masks over the 1,073 unique images and 53.77 over the 1,327 image-label tasks, which repeat multi-label images.

**Oracle@$k$.** Starting from an empty mask, the oracle greedily adds the remaining proposal that gives the greatest Dice for the current union, for at most $k$ additions. It makes $\min(k, n)$ additions, including an addition that may reduce Dice, and is recomputed for each target. It is target-aware, not deployable, and not a guaranteed global optimum.

**Template@$k$.** Training masks are resized and averaged within each class to form a learned heatmap. Its threshold preserves the median training-mask area. Each test proposal receives .7 times its IoU with this learned template plus .3 times its IoU with a fixed geometric class template, and the top $k$ proposals are joined.

**Other controls.** Area@$k$ ranks proposals by distance from the median training-mask area for the queried class. Random@3 samples without replacement and averages ten fixed draws per task. Random matched area selects a fixed random set of pixels with the class's median training area. The train class template resizes the learned heatmap to the test image and thresholds it to that area.

## H.2 Supervised Segmentation

Training annotations are split 90/10 by image ID with seed 13, so tasks from one image cannot cross the training and validation sets. Images and masks are resized to $384 \times 384$ pixels (bilinear RGB, nearest-neighbor masks). RGB is scaled to $[0, 1]$; SegFormer additionally uses ImageNet mean and standard-deviation normalization. Training uses a synchronized horizontal flip with probability .5 and AdamW (weight decay $10^{-4}$) with binary cross-entropy plus soft Dice loss. Both U-Net variants use 80 epochs, batch size 24, and learning rate $2 \times 10^{-4}$. SegFormer-B0 instead uses 40 epochs, batch size 10, and learning rate $6 \times 10^{-5}$. The checkpoint with highest validation Dice is evaluated, and probabilities are thresholded at .5.

The compact U-Net has three encoder stages, a bottleneck, three decoder stages, and base width 32. The image-only variant receives RGB. The label-conditioned variant concatenates nine tiled one-hot label channels to RGB. SegFormer-B0 initializes its encoder from `nvidia/segformer-b0-finetuned-ade-512-512`; a learned 32-dimensional label embedding is tiled over the final encoder feature map before a three-layer convolutional decoder. Reported results use one fixed seed.

## H.3 Vision-Language Model (VLM) Proposal Selection

For each image-label task, a VLM receives one rendered image containing up to the first 12 proposals from the ordered benchmark bank. Each proposal has a translucent fill, outlined boundary, and integer ID. Decoding is deterministic and uses greedy generation with at most 64 new tokens. This evaluation takes place after annotation and is unrelated to how people produced the masks.

The tested checkpoints are Qwen3-VL-32B-Instruct, Qwen3-VL-8B-Instruct, InternVL3.5-38B-Instruct, InternVL3.5-8B-Instruct, MiniCPM-V-4.5, Molmo2-8B, and LLaVA-OneVision-Qwen2-7B. The release configuration gives the corresponding public repository identifiers.

The exact proposal-selection prompt is:

> The image contains numbered SAM2.1 candidate regions. Select up to three region IDs that best mark the visible evidence supporting the composition label "{label}". Return only a JSON list of integers.

The parser uses the first bracketed substring when present, or otherwise the full response. It reads integers from left to right, keeps the first three distinct IDs that refer to rendered candidates, and ignores duplicates and out-of-range values. If no valid ID remains, or inference fails, the prediction is an empty mask. Otherwise the accepted proposal masks are joined. The invalid rate is the fraction of tasks with no accepted ID.

| Diagnostic | Value |
|---|---|
| Multi-label images | 244 |
| Different-label task pairs | 264 |
| Expert same-image, different-label Dice | .3655 |
| Expert same-image, different-label IoU | .3247 |
| Train-template same-image, different-label Dice | .3095 |
| Expert/template distance correlation | .4508 |

Table 18: Aggregate label-conditioning diagnostics.

| Label pair | Pairs | Expert D | Expert I | Template D |
|---|---|---|---|---|
| Center / Triangle | 54 | .8106 | .7674 | .5308 |
| Horizontal / Rule of thirds | 40 | .0218 | .0123 | .0000 |
| Diagonal / Horizontal | 37 | .1540 | .1224 | .3202 |
| Curved / Horizontal | 22 | .2301 | .1739 | .1401 |
| Curved / Diagonal | 18 | .3442 | .2846 | .6668 |
| Horizontal / Triangle | 12 | .1591 | .1214 | .2282 |
| Center / Rule of thirds | 10 | .7883 | .7764 | .1091 |
| Center / Symmetric | 9 | .7121 | .5880 | .5696 |
| Triangle / Vertical | 8 | .4837 | .4062 | .1188 |

Table 19: Frequent pairs of different labels assigned to the same image.

## H.4 Metrics and Confidence Intervals

For prediction $M$ and target $G$, Dice is $2|M \cap G|/(|M| + |G|)$ and IoU is $|M \cap G|/|M \cup G|$. For deployable predictions, Boundary F1 first resizes the binary prediction and target to a maximum side of 256 using bilinear filtering and a threshold of 128, extracts erosion boundaries, and accepts a match within two pixels. The historical human leave-one-out reference uses nearest-neighbor resizing before erosion boundaries. Descriptive pairwise agreement uses Pillow nearest-neighbor resizing, marks both sides of horizontal and vertical transitions without an image-frame boundary, and matches within a $5 \times 5$ square (Chebyshev distance at most two pixels). These analysis-specific conventions are preserved. Overcoverage is $|M \setminus G|/|M|$, with value zero for an empty prediction. Hit@0.5 is the fraction of tasks with Dice at least .5. Confidence intervals are the 2.5th and 97.5th percentiles of 10,000 bootstrap samples of image-label tasks, using seed 920. Tasks, rather than pixels or individual expert masks, are the sampling unit.

## H.5 Post-hoc Explanations and Deletion

The image-level recognizer is an ImageNet-initialized ResNet-50 trained on KU-PCP's nine multi-label targets. It uses an 85/15 train/validation split, 224-pixel inputs, color jitter and horizontal flips, binary cross-entropy, AdamW (learning rate $3 \times 10^{-4}$, weight decay $10^{-4}$), batch size 64, and 12 epochs. The checkpoint with best validation mean average precision (mAP) is retained. Grad-CAM uses the final ResNet stage. The DINO comparison uses `vit_small_patch16_224.dino` and cosine similarity between each patch token and the global token. Heatmaps are normalized per task. The area-matched mask preserves the expert target's area. The best-area sweep chooses the strongest target Dice from area fractions .03, .05, .08, .10, .15, .20, .30, .40, .50, .65, and .80. It is therefore an oracle threshold diagnostic rather than a deployable setting.

Deletion uses the same ResNet-50. Mean-fill replaces selected pixels with the image's per-channel mean. Blur replaces them with a Gaussian-blurred image using an odd kernel of at least 31 pixels and approximately one eighteenth of the longer image side. A flip is counted when an originally positive target prediction falls below .5 after perturbation. Full-image deletion is an upper control, not an evidence mask.

## H.6 Label Conditioning

Table 18 shows that masks for different labels on the same image have limited overlap. The pair-specific values in Table 19 show why label conditioning is not equivalent to generic foreground segmentation. Some labels can share a subject, as in center and triangle, whereas others direct attention to nearly disjoint regions in the same photograph.

| Proposal method | Dice | IoU | B-F1 | Over. | Hit |
|---|---|---|---|---|---|
| Random@3 | .0895 | .0684 | .1224 | .6449 | .0622 |
| Largest | .2491 | .2294 | .3372 | .6950 | .2419 |
| Area@1 | .3571 | .3287 | .3946 | .5352 | .3519 |
| Area@3 | .4133 | .3259 | .4250 | .5676 | .4190 |
| Template@1 | .4099 | .3724 | .4325 | .3901 | .3873 |
| Template@3 | .4147 | .3337 | .4475 | .5104 | .3986 |
| Oracle@1 | .5851 | .5197 | .5480 | .0612 | .5539 |
| Oracle@3 | .6718 | .6054 | .6317 | .0674 | .6609 |
| Oracle@5 | .6966 | .6287 | .6529 | .0755 | .6918 |
| Oracle@10 | .7040 | .6297 | .6444 | .1196 | .7189 |

Table 20: Proposal baselines against the ambiguity-aware target.

| Method | Label input | Dice | IoU | B-F1 | Over. | Hit |
|---|---|---|---|---|---|---|
| U-Net | No | .4958 | .3790 | .3071 | .4585 | .5335 |
| U-Net | Yes | .5970 | .4791 | .3904 | .3903 | .6609 |
| SegFormer-B0 | Yes | .6636 | .5762 | .4368 | .3000 | .7340 |

Table 21: Supervised segmentation against the ambiguity-aware target.

## H.7 Proposal Budgets and Supervised Models

Table 20 separates how well proposals can cover the target from how well a method selects them. The supervised comparison in Table 21 shows the benefit of supplying the queried label to U-Net and the stronger overlap achieved by SegFormer-B0. Table 22 shows that performance also varies substantially across composition classes.

## H.8 VLM Proposal Selection

The seven models in Table 23 occupy a narrow range below full-bank Template@3. Matched-budget controls are in Section J. Larger checkpoints do not consistently recover the human evidence, and invalid responses are too rare to explain the gap for most models.

## H.9 Exploratory Modality Ratings

All 8,074 annotations have a recorded modality score. Ratings are binned as segment for $r < 1/3$, mixed for $1/3 \leq r \leq 2/3$, and line for $r > 2/3$. Among the three test experts, pairwise three-bin agreement is .5386, Fleiss $\kappa = .2642$, and mean pairwise absolute rating difference is .2235. Table 24 shows that preference for regions or lines varies by composition class. Table 25 also shows variation among annotators; the training and test summaries concern different images and cannot separate annotator effects from image differences.

In Table 26, the training-set class prior exceeds every VLM because modality is strongly class-structured and the expert signal is uncertain. This experiment characterizes the metadata rather than defining a primary grounding benchmark.

## H.10 Post-hoc Explanations

The ResNet-50 classifier reaches mAP .9012, macro-F1 .8215, micro-F1 .8258, label accuracy .9541, and exact-match accuracy .6980. Table 27 shows that classifier heatmaps can overlap the human masks while agreeing less closely with their boundaries. Recognizing a label therefore does not ensure that the classifier localizes the regions people select.

## H.11 Counterfactual Deletion

Table 28 shows that human masks have much larger effects than random masks with the same area. However, full-image and overcomplete masks can cause larger drops despite weaker human alignment, and Template@3 exceeds the higher-Dice Oracle@3. Deletion therefore supports functional relevance while measuring something different from agreement with human evidence.

| Class | U-Net, no label | U-Net + label | SegFormer-B0 |
|---|---|---|---|
| Horizontal | .2838 | .4705 | .6459 |
| Vertical | .4465 | .5264 | .5281 |
| Diagonal | .3719 | .4596 | .4531 |
| Curved | .3789 | .4452 | .5052 |
| Triangle | .5889 | .6595 | .7841 |
| Rule of thirds | .6397 | .7840 | .8704 |
| Center | .6133 | .6556 | .7365 |
| Symmetric | .5900 | .8059 | .7770 |
| Pattern | .5337 | .5705 | .4513 |

Table 22: Per-class Dice for supervised models.

| Model | Dice | IoU | B-F1 | Over. | Hit | Invalid |
|---|---|---|---|---|---|---|
| Qwen3-VL-32B | .3946 | .3243 | .4396 | .5231 | .3760 | .0008 |
| Qwen3-VL-8B | .3770 | .3111 | .4309 | .5622 | .3519 | .0008 |
| MiniCPM-V 4.5 | .3693 | .3081 | .4107 | .5435 | .3429 | .0000 |
| InternVL3.5-38B | .3634 | .3003 | .4114 | .5408 | .3384 | .0211 |
| LLaVA-OneVision-7B | .3312 | .2579 | .3971 | .6608 | .3120 | .0000 |
| Molmo2-8B | .3286 | .2747 | .3909 | .6071 | .3044 | .0000 |
| InternVL3.5-8B | .2910 | .2416 | .3618 | .6124 | .2668 | .0580 |

Table 23: VLM selection of up to three numbered proposals.

| Class | Train $n$ | Mean $r$ | S / M / L (%) | Test $n$ | Mean $r$ | S / M / L (%) |
|---|---|---|---|---|---|---|
| Horizontal | 758 | .47 | 13 / 83 / 4 | 633 | .55 | 9 / 64 / 27 |
| Vertical | 195 | .11 | 84 / 15 / 1 | 198 | .22 | 70 / 24 / 7 |
| Diagonal | 471 | .44 | 28 / 60 / 12 | 567 | .59 | 12 / 45 / 43 |
| Curved | 235 | .01 | 99 / 1 / 0 | 276 | .14 | 84 / 13 / 3 |
| Triangle | 374 | .34 | 45 / 52 / 4 | 513 | .54 | 22 / 38 / 40 |
| Rule of thirds | 923 | .02 | 98 / 1 / 0 | 375 | .20 | 72 / 22 / 5 |
| Center | 750 | .03 | 98 / 2 / 0 | 933 | .21 | 71 / 24 / 5 |
| Symmetric | 232 | .22 | 72 / 25 / 3 | 297 | .29 | 55 / 37 / 8 |
| Pattern | 155 | .01 | 100 / 0 / 0 | 189 | .17 | 77 / 20 / 3 |

Table 24: Exploratory modality ratings for both splits. S/M/L denote segment/mixed/line percentages. Training has one rating per task and test has three. The splits contain different images and annotators, so this is an unpaired descriptive comparison. The main figure reports expert-test ratings.

| Rater | Split | Ratings | Mean $r$ | Segment | Mixed | Line |
|---|---|---|---|---|---|---|
| rater_01 | Test | 1327 | .250 | 58.9% | 33.2% | 7.9% |
| rater_02 | Train | 4093 | .196 | 67.5% | 29.8% | 2.8% |
| rater_03 | Test | 1327 | .495 | 34.5% | 27.4% | 38.1% |
| rater_04 | Test | 1327 | .337 | 45.9% | 44.6% | 9.5% |

Table 25: Modality scores for each completed annotator. These summaries describe recorded scores; they are not evidence of agreement between the training annotator and test experts, who worked on different image splits.

| Modality predictor | Accuracy | Macro-F1 | MAE | Invalid |
|---|---|---|---|---|
| Always segment | .4943 | .2205 | .3082 | – |
| Training-set class prior | .7204 | .5008 | .1398 | – |
| Molmo2-8B | .4348 | .3522 | .2833 | .0000 |
| Qwen3-VL-32B | .3338 | .2832 | .4668 | .0000 |
| Qwen3-VL-8B | .3157 | .3048 | .4672 | .0000 |
| LLaVA-OneVision-7B | .3263 | .2615 | .4759 | .0000 |
| InternVL3.5-38B | .2976 | .2724 | .4048 | .0023 |
| InternVL3.5-8B | .3090 | .2776 | .4555 | .0000 |
| MiniCPM-V 4.5 | .1959 | .1870 | .5629 | .0000 |

Table 26: VLM diagnostics for exploratory modality metadata. MAE is mean absolute error.

| Method | Dice | IoU | B-F1 | Over. | Hit |
|---|---|---|---|---|---|
| DINO, area matched | .2102 | .1432 | .1413 | .7898 | .1515 |
| DINO, best area sweep | .3404 | .2314 | .2018 | .7307 | .2524 |
| Grad-CAM, area matched | .5176 | .3793 | .1371 | .4907 | .6074 |
| Grad-CAM, best area sweep | .5833 | .4328 | .1947 | .4927 | .7242 |

Table 27: Classifier-derived explanation masks.

| Mask source | Blur drop | Fill drop | Blur flip | Fill flip | Mask D | Area |
|---|---|---|---|---|---|---|
| Human consensus | 1.0192 | 2.0311 | .0965 | .2095 | 1.0000 | .2827 |
| Random matched area | .1645 | .3754 | .0173 | .0347 | .2827 | .2827 |
| Edge matched area | 1.0981 | 2.2527 | .0942 | .2306 | .4499 | .3246 |
| Full image | 2.3069 | 4.0795 | .2622 | .7611 | .4085 | 1.0000 |
| U-Net + label | 1.1649 | 2.8026 | .1168 | .2833 | .5970 | .3160 |
| SAM2.1 Area@3 | .5675 | 1.5486 | .0603 | .1703 | .4133 | .3294 |
| SAM2.1 Oracle@3 | .5531 | 1.3079 | .0565 | .1417 | .6718 | .1733 |
| SAM2.1 Template@3 | .7084 | 1.7485 | .0701 | .1922 | .4147 | .3899 |
| Qwen3-VL-32B | .6249 | 1.5760 | .0610 | .1726 | .3946 | .3426 |

Table 28: Classifier logit drops and prediction flips after deleting each mask. Mask D is Dice to the human target.

| Path | Contents |
|---|---|
| annotations/single_annotator_train/ | 4,093 records from 3,169 train images and their binary masks |
| annotations/expert_test/expert_01..03/ | Three pseudonymous bundles of 1,327 records and masks |
| consensus/strict/ | 1,327 strict-majority masks |
| consensus/ambiguity_aware/ | 1,327 ambiguity-aware masks |
| consensus/manifest.csv | Task-to-consensus-mask index |
| metadata/preference_ratings.csv | Exploratory expert modality ratings |
| metadata/process/ | Scalar workflow labels and aggregate summaries |
| split_manifest.csv | Split-local image, task, and annotator counts |
| schemas/ | Annotation and workflow field definitions |
| benchmark_proposal_config.json | Fixed proposal benchmark settings |
| file_manifest.csv, checksums.csv | Complete inventory and SHA-256 checksums |
| README.md, DATASET_CARD.md, DATASHEET.md | Scope, construction, uses, and limitations |
| LICENSE.md, CITATION.cff | Annotation-layer license and citation information |
| RECONSTRUCTION.md, evaluate_smoke_test.py | KU-PCP join instructions and integrity test |

Table 29: Annotation assets for the evaluator. The dataset download described below contains the complete annotations and metadata for all four annotators.

# I Release and Reconstruction

KU-PCPE contains positive image-label grounding tasks for nine KU-PCP composition classes. A binary mask marks visible regions selected as support for a label already judged applicable. The training split contains one annotator; the official test split contains three experts represented by pseudonyms and both strict and ambiguity-aware consensus targets. The masks are subjective judgments of photographic structure, not identity or sensitive-attribute labels. Known limitations are candidate-assisted collection, one-annotator training supervision, positive-only queries, and exploratory modality ratings with modest agreement. Table 29 lists the annotation assets used by the evaluator: benchmark masks, derived consensus masks, metadata, and validation files. These supporting assets allow individual expert judgments and both consensus rules to be evaluated separately.

The dataset download, kupcp_completed_public_20261023_v1.tar.gz, contains the complete annotations and metadata for the training annotator and all three test experts. Every annotation has a segment-versus-line score, recorded settings, and action counts. It also includes final and component masks, a pseudonymous participant index, summaries by annotator, class, and task, and relative tool-event timelines. The separate reproducibility project includes a script that joins these records to the fixed per-task benchmark scores and regenerates all-rater workflow tables, including confidence intervals and per-rater sensitivity results.

To reconstruct the benchmark, users obtain KU-PCP through its original distribution and set KUPCP_ROOT to a directory containing train_img/ and test_img/. Training records resolve under the former and expert-test records under the latter. The filename namespaces remain separate. From the release root, the verification command is:

```
KUPCP_ROOT=/path/to/KU-PCP python -I evaluate_smoke_test.py
```

The expected final line is:

```
SMOKE TEST PASSED: 4093 train tasks; 1327 expert-test tasks; identity Dice=1.000000
```

The integrity test decodes every annotation PNG, verifies binary values and foreground areas, measures pixel-exact round-trip Dice, and reconstructs both consensus targets. It also checks every package checksum, mask path, consensus link, task identity, PNG dimension, split count, and referenced KU-PCP image. It was run from a fresh copy of the release in an isolated Python process. Source photographs are not redistributed or relicensed. The KU-PCPE annotation layer is licensed under CC BY 4.0; users must separately follow KU-PCP's original terms.

In the complete export, tool_usage_by_task.csv gives tool counts for each annotation; summaries by annotator and class are supplied separately. tool_timeline.csv groups recorded events into five-second bins within observed clock segments, with time values capped at 30 minutes. Original participant identifiers, calendar timestamps, coordinates, free text, and raw event payloads are withheld. These records describe observed interface use, not exact inference calls or active working time. Discarded drafts and unretained revision histories are not reconstructed.

# J Matched 12-Proposal Controls

We replay the exact original first 12 masks in shard order and their 1-based IDs, without regenerating or re-ranking proposals. All 1,073 image overlays match the original rendering byte for byte. The 67 tasks (59 distinct images) with fewer proposals use their available bank. Saved responses from all seven VLMs are evaluated with the parser described in Section H. These are saved-prediction re-evaluations, not new VLM inference runs.

The template scores and training masks are unchanged. Greedy selection makes $\min(3, n)$ additions even when a later addition reduces Dice, resolving ties by original candidate order. The exact reference enumerates all subsets with exactly $\min(3, n)$ masks, and separately all subsets of at most three masks, including the empty subset. Each oracle optimizes Dice separately for each target; predictions are never inverted. Other reported metrics evaluate the Dice-maximizing subset and are not independently optimized ceilings. Exact integer subset enumeration passed 156 randomized and empty-case checks against explicit binary unions. Bootstrap intervals resample image-label tasks 10,000 times with seed 920; repeated images are not independent resampling clusters, so these intervals should not be read as image-cluster uncertainty.

Under both consensus rules, the template and strongest VLM have similar Dice, while the exact references recover more of the target from the same candidates (Tables 30 to 33).

| Method | Dice | IoU | B-F1 |
|---|---|---|---|
| Template@3 | 0.4016 [0.3831, 0.4200] | 0.3204 [0.3030, 0.3382] | 0.4412 [0.4255, 0.4567] |
| Qwen3-VL-32B | 0.3946 [0.3752, 0.4139] | 0.3243 [0.3057, 0.3431] | 0.4396 [0.4232, 0.4556] |
| Greedy Oracle@3 | 0.6400 [0.6202, 0.6595] | 0.5745 [0.5539, 0.5952] | 0.5963 [0.5777, 0.6154] |
| Exact, exactly 3 | 0.6401 [0.6202, 0.6598] | 0.5746 [0.5534, 0.5958] | 0.5964 [0.5771, 0.6152] |
| Exact, up to 3 | 0.6448 [0.6248, 0.6644] | 0.5820 [0.5604, 0.6030] | 0.6096 [0.5896, 0.6292] |
| InternVL3.5-38B | 0.3634 [0.3438, 0.3832] | 0.3003 [0.2817, 0.3194] | 0.4114 [0.3951, 0.4282] |
| InternVL3.5-8B | 0.2910 [0.2724, 0.3103] | 0.2416 [0.2240, 0.2597] | 0.3618 [0.3450, 0.3785] |
| Qwen3-VL-8B | 0.3770 [0.3568, 0.3964] | 0.3111 [0.2917, 0.3298] | 0.4309 [0.4148, 0.4469] |
| Molmo2-8B | 0.3286 [0.3085, 0.3489] | 0.2747 [0.2557, 0.2941] | 0.3909 [0.3743, 0.4075] |
| LLaVA-OneVision-7B | 0.3312 [0.3134, 0.3492] | 0.2579 [0.2416, 0.2743] | 0.3971 [0.3825, 0.4119] |
| MiniCPM-V-4.5 | 0.3693 [0.3490, 0.3897] | 0.3081 [0.2888, 0.3276] | 0.4107 [0.3937, 0.4279] |

Table 30: Original shared-candidate controls, ambiguity-aware consensus. Means and 95% task-bootstrap intervals; all rows use 1,327 tasks.

| Method | Overcoverage | Hit@0.5 | Invalid |
|---|---|---|---|
| Template@3 | 0.5401 [0.5195, 0.5605] | 0.3828 [0.3572, 0.4092] | 0.0000 [0.0000, 0.0000] |
| Qwen3-VL-32B | 0.5231 [0.5015, 0.5446] | 0.3760 [0.3497, 0.4024] | 0.0008 [0.0000, 0.0023] |
| Greedy Oracle@3 | 0.1226 [0.1087, 0.1371] | 0.6330 [0.6074, 0.6594] | 0.0000 [0.0000, 0.0000] |
| Exact, exactly 3 | 0.1214 [0.1074, 0.1361] | 0.6330 [0.6066, 0.6586] | 0.0000 [0.0000, 0.0000] |
| Exact, up to 3 | 0.0636 [0.0535, 0.0742] | 0.6368 [0.6104, 0.6631] | 0.0000 [0.0000, 0.0000] |
| InternVL3.5-38B | 0.5408 [0.5191, 0.5627] | 0.3384 [0.3127, 0.3640] | 0.0211 [0.0136, 0.0294] |
| InternVL3.5-8B | 0.6124 [0.5902, 0.6345] | 0.2668 [0.2434, 0.2901] | 0.0580 [0.0460, 0.0708] |
| Qwen3-VL-8B | 0.5622 [0.5406, 0.5838] | 0.3519 [0.3255, 0.3775] | 0.0008 [0.0000, 0.0023] |
| Molmo2-8B | 0.6071 [0.5853, 0.6291] | 0.3044 [0.2796, 0.3293] | 0.0000 [0.0000, 0.0000] |
| LLaVA-OneVision-7B | 0.6608 [0.6419, 0.6791] | 0.3120 [0.2871, 0.3369] | 0.0000 [0.0000, 0.0000] |
| MiniCPM-V-4.5 | 0.5435 [0.5213, 0.5659] | 0.3429 [0.3173, 0.3693] | 0.0000 [0.0000, 0.0000] |

Table 31: Original shared-candidate controls, ambiguity-aware consensus. Means and 95% task-bootstrap intervals; all rows use 1,327 tasks.

| Method | Dice | IoU | B-F1 |
|---|---|---|---|
| Template@3 | 0.3989 [0.3801, 0.4176] | 0.3201 [0.3023, 0.3380] | 0.4348 [0.4190, 0.4506] |
| Qwen3-VL-32B | 0.3988 [0.3792, 0.4185] | 0.3291 [0.3103, 0.3481] | 0.4383 [0.4219, 0.4549] |
| Greedy Oracle@3 | 0.6431 [0.6232, 0.6630] | 0.5816 [0.5605, 0.6028] | 0.6080 [0.5886, 0.6270] |
| Exact, exactly 3 | 0.6432 [0.6228, 0.6628] | 0.5818 [0.5602, 0.6026] | 0.6085 [0.5890, 0.6275] |
| Exact, up to 3 | 0.6498 [0.6295, 0.6703] | 0.5913 [0.5696, 0.6132] | 0.6253 [0.6052, 0.6456] |
| InternVL3.5-38B | 0.3674 [0.3476, 0.3872] | 0.3059 [0.2871, 0.3249] | 0.4085 [0.3915, 0.4256] |
| InternVL3.5-8B | 0.2938 [0.2740, 0.3138] | 0.2465 [0.2281, 0.2657] | 0.3560 [0.3386, 0.3738] |
| Qwen3-VL-8B | 0.3798 [0.3597, 0.4008] | 0.3155 [0.2964, 0.3353] | 0.4279 [0.4113, 0.4451] |
| Molmo2-8B | 0.3296 [0.3092, 0.3504] | 0.2781 [0.2586, 0.2978] | 0.3852 [0.3681, 0.4026] |
| LLaVA-OneVision-7B | 0.3334 [0.3155, 0.3519] | 0.2615 [0.2452, 0.2782] | 0.3911 [0.3761, 0.4065] |
| MiniCPM-V-4.5 | 0.3686 [0.3485, 0.3888] | 0.3095 [0.2901, 0.3288] | 0.4063 [0.3891, 0.4237] |

Table 32: Original shared-candidate controls, strict consensus. Means and 95% task-bootstrap intervals; all rows use 1,327 tasks.

| Method | Overcoverage | Hit@0.5 | Invalid |
|---|---|---|---|
| Template@3 | 0.5613 [0.5410, 0.5813] | 0.3791 [0.3527, 0.4047] | 0.0000 [0.0000, 0.0000] |
| Qwen3-VL-32B | 0.5380 [0.5166, 0.5592] | 0.3828 [0.3564, 0.4099] | 0.0008 [0.0000, 0.0023] |
| Greedy Oracle@3 | 0.1492 [0.1334, 0.1650] | 0.6383 [0.6127, 0.6639] | 0.0000 [0.0000, 0.0000] |
| Exact, exactly 3 | 0.1486 [0.1325, 0.1646] | 0.6383 [0.6119, 0.6639] | 0.0000 [0.0000, 0.0000] |
| Exact, up to 3 | 0.0779 [0.0663, 0.0898] | 0.6436 [0.6172, 0.6692] | 0.0000 [0.0000, 0.0000] |
| InternVL3.5-38B | 0.5522 [0.5305, 0.5736] | 0.3482 [0.3225, 0.3730] | 0.0211 [0.0136, 0.0294] |
| InternVL3.5-8B | 0.6231 [0.6010, 0.6452] | 0.2735 [0.2502, 0.2984] | 0.0580 [0.0460, 0.0708] |
| Qwen3-VL-8B | 0.5812 [0.5592, 0.6030] | 0.3602 [0.3346, 0.3866] | 0.0008 [0.0000, 0.0023] |
| Molmo2-8B | 0.6197 [0.5975, 0.6416] | 0.3067 [0.2826, 0.3316] | 0.0000 [0.0000, 0.0000] |
| LLaVA-OneVision-7B | 0.6713 [0.6527, 0.6896] | 0.3157 [0.2909, 0.3406] | 0.0000 [0.0000, 0.0000] |
| MiniCPM-V-4.5 | 0.5696 [0.5480, 0.5914] | 0.3474 [0.3225, 0.3723] | 0.0000 [0.0000, 0.0000] |

Table 33: Original shared-candidate controls, strict consensus. Means and 95% task-bootstrap intervals; all rows use 1,327 tasks.

**Paired comparisons.** On the ambiguity-aware target, Qwen3-VL-32B minus Template@3 (12) has mean Dice $-.00697$ with 95% interval $[-.02157, .00753]$; under strict consensus, the difference is $-.00006$ with interval $[-.01450, .01435]$. Neither supports a Dice advantage over the matched template. Full-bank minus 12-bank greedy Dice is .03173 $[.02678, .03703]$; the template difference is .01317 $[.00824, .01828]$. Exactly-three exact minus greedy Dice is .00008 $[.00004, .00014]$, and allowing up to three improves the exact reference by .00468 $[.00361, .00596]$. These isolate budget and selection effects rather than changing the learned predictor. Complete paired intervals for all five metrics are included with the result files. Class results (Tables 34 and 35) show that no deployable selector leads throughout, while oracle coverage varies substantially with composition.

Table 34: Per-class Dice on the original 12 candidates, ambiguity-aware consensus. H: horizontal; V: vertical; D: diagonal; C: curved; T: triangle; R: rule of thirds; Ce: center; S: symmetric; P: pattern.

| Method | H | V | D | C | T | R | Ce | S | P |
|---|---|---|---|---|---|---|---|---|---|
| Template@3 | 0.566 | 0.340 | 0.354 | 0.291 | 0.408 | 0.334 | 0.457 | 0.307 | 0.213 |
| Qwen3-VL-32B | 0.571 | 0.544 | 0.356 | 0.281 | 0.301 | 0.363 | 0.437 | 0.302 | 0.184 |
| Greedy Oracle@3 | 0.839 | 0.716 | 0.581 | 0.509 | 0.563 | 0.858 | 0.655 | 0.447 | 0.266 |
| Exact, exactly 3 | 0.840 | 0.716 | 0.581 | 0.509 | 0.563 | 0.858 | 0.655 | 0.447 | 0.266 |
| Exact, up to 3 | 0.842 | 0.717 | 0.585 | 0.514 | 0.567 | 0.865 | 0.664 | 0.447 | 0.267 |
| InternVL3.5-38B | 0.552 | 0.376 | 0.339 | 0.302 | 0.282 | 0.236 | 0.429 | 0.281 | 0.161 |
| InternVL3.5-8B | 0.517 | 0.245 | 0.294 | 0.248 | 0.199 | 0.186 | 0.311 | 0.188 | 0.158 |
| Qwen3-VL-8B | 0.590 | 0.436 | 0.367 | 0.300 | 0.312 | 0.319 | 0.378 | 0.281 | 0.181 |
| Molmo2-8B | 0.556 | 0.285 | 0.276 | 0.262 | 0.233 | 0.270 | 0.391 | 0.186 | 0.156 |
| LLaVA-OneVision-7B | 0.548 | 0.295 | 0.331 | 0.251 | 0.262 | 0.338 | 0.313 | 0.247 | 0.159 |
| MiniCPM-V-4.5 | 0.543 | 0.398 | 0.351 | 0.278 | 0.270 | 0.398 | 0.395 | 0.279 | 0.173 |

Table 35: Per-class Dice on the original 12 candidates, strict consensus. H: horizontal; V: vertical; D: diagonal; C: curved; T: triangle; R: rule of thirds; Ce: center; S: symmetric; P: pattern.

| Method | H | V | D | C | T | R | Ce | S | P |
|---|---|---|---|---|---|---|---|---|---|
| Template@3 | 0.566 | 0.357 | 0.327 | 0.277 | 0.406 | 0.339 | 0.460 | 0.309 | 0.217 |
| Qwen3-VL-32B | 0.568 | 0.557 | 0.366 | 0.279 | 0.302 | 0.368 | 0.443 | 0.298 | 0.207 |
| Greedy Oracle@3 | 0.840 | 0.718 | 0.588 | 0.474 | 0.569 | 0.865 | 0.660 | 0.448 | 0.296 |
| Exact, exactly 3 | 0.840 | 0.718 | 0.588 | 0.474 | 0.570 | 0.866 | 0.661 | 0.448 | 0.296 |
| Exact, up to 3 | 0.843 | 0.721 | 0.599 | 0.490 | 0.576 | 0.873 | 0.669 | 0.448 | 0.296 |
| InternVL3.5-38B | 0.553 | 0.382 | 0.348 | 0.291 | 0.285 | 0.241 | 0.436 | 0.278 | 0.177 |
| InternVL3.5-8B | 0.525 | 0.248 | 0.303 | 0.210 | 0.203 | 0.191 | 0.315 | 0.192 | 0.166 |
| Qwen3-VL-8B | 0.584 | 0.438 | 0.383 | 0.283 | 0.320 | 0.324 | 0.381 | 0.278 | 0.193 |
| Molmo2-8B | 0.559 | 0.289 | 0.286 | 0.218 | 0.236 | 0.275 | 0.395 | 0.188 | 0.163 |
| LLaVA-OneVision-7B | 0.549 | 0.302 | 0.336 | 0.226 | 0.266 | 0.343 | 0.316 | 0.244 | 0.183 |
| MiniCPM-V-4.5 | 0.534 | 0.421 | 0.342 | 0.248 | 0.272 | 0.403 | 0.401 | 0.276 | 0.195 |